

TCL Assignment 1 Report

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1 Introduction

Tool Command Language (TCL) is a powerful scripting language commonly utilized in the semiconductor industry for automation, simulation setup, and design flow control. This report documents the implementation of basic logic, bitwise arithmetic, and mathematical operations executed using TCL scripts.

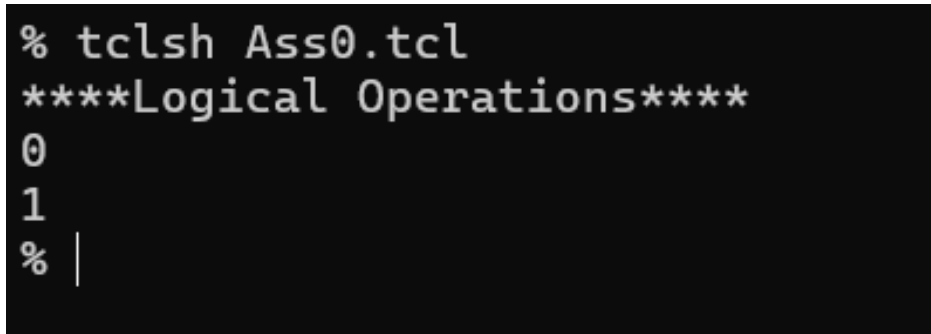
2 Assignment 1.0: Logical Operations

This script, `Ass0.tcl`, initializes variables and exercises logical operations. In TCL, logical operations return 1 for true and 0 for false.

2.1 Code Snippet

```
1 puts "****Logical Operations****"
2
3 set a 5
4 set b -1
5 set c 0
6
7 set var0 [expr {$a && $c}]
8 set var1 [expr {$a || $b}]
9
10 puts $var0
11 puts $var1
```

2.2 Terminal Output

A terminal window with a black background and white text. The text shows the execution of a Tcl script named Ass0.tcl. The output consists of the header '****Logical Operations****', followed by the number '0', then '1', and finally a '%' character followed by a vertical bar '|'.

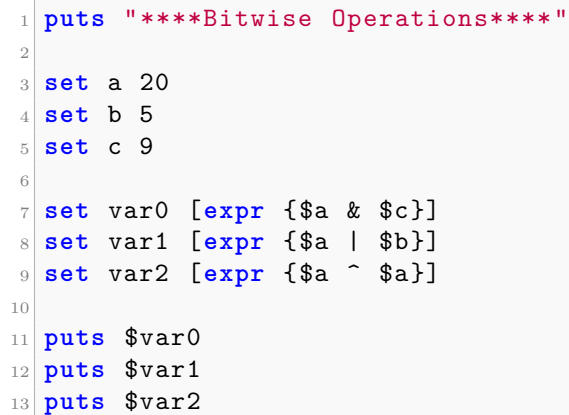
```
% tclsh Ass0.tcl
****Logical Operations****
0
1
% |
```

Figure 1: Output of Logical Operations script

3 Assignment 1.1: Bitwise Operations

The script `Ass1.tcl` evaluates bitwise operations between values. The bitwise operators performing logic gates bit-by-bit are AND (`&`), OR (`|`), and XOR (`^`).

3.1 Code Snippet

A code editor window with a light gray background. It contains a Tcl script snippet for bitwise operations. The script starts with a header line, followed by three 'set' commands to initialize variables a, b, and c. Then, it uses 'set' with 'expr' to calculate the bitwise AND, OR, and XOR of a and b, storing the results in var0, var1, and var2 respectively. Finally, it uses three 'puts' commands to print the values of var0, var1, and var2.

```
1 puts "****Bitwise Operations****"
2
3 set a 20
4 set b 5
5 set c 9
6
7 set var0 [expr {$a & $c}]
8 set var1 [expr {$a | $b}]
9 set var2 [expr {$a ^ $a}]
10
11 puts $var0
12 puts $var1
13 puts $var2
```

3.2 Terminal Output

```
% tclsh Ass1.tcl
****Bitwise Operations****
0
21
0
% |
```

Figure 2: Output of Bitwise Operations script

4 Assignment 1.2: Mathematical Operations

The script Ass2.tcl utilizes Tcl math functions `abs()` and `sqrt()` to perform calculations.

4.1 Code Snippet

```
1 puts "****Mathematical Operations****"
2
3 set a -64
4 set b 49
5
6 set var0 [expr {abs($a)}]
7 set var1 [expr {sqrt($b)}]
8
9 puts $var0
10 puts $var1
```

4.2 Terminal Output

```
% tclsh Ass2.tcl
****Mathematical Operations****
64
7.0
% |
```

Figure 3: Output of Mathematical Operations script